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# Evaluation of the Diagnostic and Predictive Value of Serum Levels of ANT1, ATG5, and Parkin in Multiple Sclerosis

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ABSTRACT

*Objective:* Multiple Sclerosis (MS) is a disease of the central nervous system, which ultimately may lead to various disabilities in patients. No definitive cure has yet been developed for the disease. MRI is the method of choice for imaging MS plaques, which would be useful in disease diagnosis as it becomes progressive. Therefore, this study aimed to investigate the serum levels of ANT1 (adenine nucleotide translocase 1), ATG5 (autophagy-related protein 5), and Parkin in patients with MS, all of which play essential roles in MS pathophysiology, as novel serum biomarkers for early diagnosis of the disease.

*Design and methods:* Forty patients in the early stages of the disease, and 40 healthy individuals were selected as the case and control groups. Upon sampling, the serum levels of the biomarkers were measured.

*Results:* The results indicated that autophagy, mitophagy, and mitochondrial apoptosis were different in the case and control groups. The oxidative stress level evaluation revealed low concertation of total antioxidant status (TAS) in the MS patients, while a partial increase accompanied the malondialdehyde (MDA). No significant correlation was observed between oxidative stress and autophagy or mitophagy factors.

*Conclusion:* According to the results obtained from this study, the evaluation of serum levels of ANT1, ATG5, and Parkin could be applied in the diagnosis and follow-up of MS patients.

## 1. Introduction

Multiple Sclerosis (MS) is a chronic inflammatory disease of the central nervous system. MS patients show varying degrees of physical and neurological disorders. Moreover, multiple sclerosis is considered as one of the most common neurodegenerative diseases that can cause many neurological symptoms, including loss of tactile sensation, visual impairment, double vision, muscle weakness, and balance and coordination difficulties. Vitamin D deficiency, smoking, childhood obesity, infectious agents, such as Epstein - Barr virus (EBV), are perceived to contribute to the development of this disease. MS is twice as common in women as in men [1–3].

It is estimated that MS is most prevalent in the North American and European countries, while East Asian countries have the lowest number of cases [1,4]. An average of 34.2 cases per 100,000 individuals is affected by multiple sclerosis in Iran. Previous studies in the country revealed that about 44.8 women and 16.5 men per 100,000 are suffering from MS [5,6].

MS is usually diagnosed based on clinical symptoms. Also, an MRI brain scan can be used as a diagnostic method for advanced stages of the disease, known as the primary progressive multiple sclerosis (PPMS). No reliable diagnostic test has been developed as of yet, and treatment options are only useful in the early stages of the disease, i.e., the relapsing-remitting multiple sclerosis (RRMS). Thus, most of the MS studies have primarily focused on this type of disease [7]. Moreover, genetic and environmental factors contribute to MS progression, in addition to oxidative stress, loss of mitochondrial activity, and autoimmune disorders, which all have significant roles in the pathology of

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numerous neurodegenerative diseases, including MS [8].

Cells of the immune system such as macrophages, microglia, and astrocytes produce neurotoxic molecules such as reactive oxygen species (ROS), reactive nitrogen species (RNS), and Nitric Oxide (NO). On the other hand, mitochondria and its DNA are highly susceptible to oxidative damage. Parkin is a ligase that plays a crucial role in mitophagy, and suppression of reactive oxygen species. Parkin targets the damaged mitochondria leading to mitophagy. This process is critical in the removal of oxygen radicals and other ROS sources [9].

The rate of autophagy is usually kept at a low level in a living cell; however, this rate is increased under some conditions, such as a shortage in nutrient supply, organ failure, oxidative stress, and the presence of chronic stimuli. Moreover, the absence of autophagy-related protein 5 (ATG5), a protein involved in autophagy, can lead to autophagy suppression [10]. The Immune system is heavily involved in the autophagy process, namely in pathogen recognition and antigen presentation. A high rate of Autophagy in MS patients can activate the pro-survival signal for T-cells [11].

Adenine Nucleotide Translocase 1 (ANT1) is the most abundant protein in the inner mitochondrial membrane. It is essential for the stability of ATP synthase and mitochondrial membrane potential. ANT1 can modulate the mitochondrial apoptosis pathway, where pro-apoptotic and anti-apoptotic molecules affect pathways of signal transduction [12].

The present study aimed to investigate the relationship between autophagy and mitophagy factors, ANT1, and oxidative stress in MS patients. Finding a correlation among these factors could be beneficial in the early detection and control of the disease. Furthermore, it can lead to improvements in the treatment, prognosis, and understanding of the underlying mechanism of the disease.

## 2. Materials and methods

### 2.1. Subject recruitment

All participants received a full explanation of the aims of the study and were offered informed consent forms. Ethical clearance was obtained from the Ethics Committee of Tabriz University of Medical Sciences (IR.TBZMED.REC.1398.312). This case-control study comprised 40 patients with RRMS, who fulfilled the revised McDonald criteria [13], and 40 healthy individuals. Patients were neurologically examined by MS neurologists where smoking, alcohol consumption, taking immunosuppressive drugs, and using blood lipid-lowering agents were considered as exclusion criteria Disability was scored by the Expanded Disability Status Scale (EDSS), and the Multiple Sclerosis Severity Score scored the disease severity. A relapse was defined as an episode of neurological disturbance lasting for at least 24 hours, which could not be explained by another cause. We used a standard MRI protocol for multiple sclerosis with IV gadolinium (Gd) as the contrast. The disease activity was defined as a relapse or Gd-enhancing lesion. Due to the lack of data for T2 lesions in MRI, we used a modified no evidence of disease activity (NEDA): [11] the absence of contrast-enhancing lesions on MRI, absence of confirmed disability progression defined as the absence of increased post-treatment disability by 1.0 if the EDSS score was0 to 5.5 at baseline or by 0.5 if the EDSS score was $\geq$ 6.0 at baseline, and lack of relapses.[14,12]

### 2.2. Sample preparation

Ten ml of the blood sample was withdrawn from each individual after an overnight fasting period. Blood samples were centrifuged at 3,000 rpm for 4 minutes at 4˚C and stored at -80˚C. Finally, after 15 minutes of centrifugation at 3000 rpm and ambient temperature, the sera were isolated and kept at -70˚C for subsequent measurement of biochemical indicators.

### 2.3. Measurement of lipid profile

#### 2.3.1. Measurement of triglyceride concentration

The Pars Azmoon triglyceride assay kit (GPO-PAP, Cat. No. 132 504 H917) was used to measure triglyceride in an automated biochemistry analyzer (Hitachi 917). Glycerol was separated from fatty acids utilizing the lipoprotein lipase. Then, the hydrogen peroxide, released from glycerol reaction with 4-amino antipyrine and phenol, formed a quinone imine, which was catalyzed in the presence of peroxidase. The amount of quinone imine is directly proportional to the amount of triglyceride and can be measured photometrically.

#### 2.3.2. Measurement of cholesterol concentration

The Pars Azmoon assay kit (Cat. No. 110 500 BT) was used to measure cholesterol in an automatic biochemistry analyzer (BT3000). Moreover, the hydrogen peroxide, formed by the hydrolysis and oxidation of cholesterol, reacted with phenol and 4-amino antipyrine in the presence of peroxidase and formed quinone imine. The amount of quinone imine is directly proportional to the amount of cholesterol and can be measured photometrically.

#### 2.3.3. Measurement of high-density lipoprotein (HDL) concentration

The Pars Azmoon assay kit (Order No. 1050012) was utilized to measure the amount of HDL. The inhibition of chylomicron, VLDL, and LDL by antibodies was the primary method for enzymatic HDL concentration measurement.

### 2.4. Measurement of malondialdehyde concentration

The Malondialdehyde colorimetric assay kit (USA, Cat. No. E-BC-K025-S) and the TBA method were used to measure MDA. In this assay, the free MDA present in the sample reacts with thiobarbituric acid (TBA) and produces an MDA-TBA adduct, which can easily be quantified colorimetrically (OD: 532 nm).

### 2.5. Measurement of total antioxidant status concentration

The Randox Total Antioxidant Status kit (UK, Cat. No. NX 2332) was utilized to measure the samples' full antioxidant capacity. The peroxidase substrate, namely, the 2,2′-Azino-di-(3-ethylbenzothiazoline sulfonic acid) or ABTS, was incubated with H2O2 and metmyoglobin to generate the radical cation form of ABTS (ABTS+). The ABTS + had a solid blue-green color, which was measured at a wavelength of 600 nm. The antioxidants present in the samples suppressed the color formation to a degree proportional to their concentrations.

### 2.6. Measurement of autophagy and mitophagy factors

#### 2.6.1. Measurement of ATG5 concentration

MyBioSource 96-strip-wells ELISA kit (Cat No. MBS7209535) was used to measure ATG5 in samples. The sample and buffer were incubated together with the ATG5-HRP conjugate in a pre-coated plate for one hour. The wells were decanted and washed five times once the incubation period was finished. The wells were then incubated with a substrate for HRP enzyme. The product of the enzyme-substrate reaction formed a blue-colored complex. Finally, a stop solution was added to stop the result, which turned the solution's color into yellow. The intensity of the color was measured spectrophotometrically at 450 nm in a microplate reader.

#### 2.6.2. Measurement of Parkin concentration

MyBioSource ELISA kit (Cat. No. MBS732278) was utilized to measure Parkin's level in samples. Moreover, the competitive enzyme immunoassay technique was applied while using a polyclonal anti-Parkin antibody and a Parkin-HRP conjugate.

#### 2.6.3. Measurement of ANT1 concentration

Fine Test ELISA kit (Cat. No. EH1721) was used to measure the level of ANT1 in samples. The governing principle was based on standard sandwich enzyme-linked immunosorbent assay technology, where a biotin-conjugated antibody was used as the detection antibody. The standards, test samples, and biotin-conjugated detection antibody were added to the wells subsequently and rinsed with wash buffer. HRP-Streptavidin was then added, and unbound conjugates were rinsed with wash buffer. TMB substrates were used to visualize HRP enzymatic reaction, where TMB was catalyzed by HRP to produce a blue color product that changed into yellow after adding the acidic stop solution. The density of yellow complex is proportional to the target amount of sample captured in plate.

### 2.7. Statistical analysis

Data analysis was performed using SPSS Statistics V22.0 and GraphPad Prism 8. Pearson's correlation coefficient was used to evaluate the correlations between the serum concentrations of ANT1, Parkin, and ATG5 with the oxidative stress markers in patients with multiple sclerosis and control groups. A significant difference between the study groups was defined by $p<0.05$.

## 3. Results

### 3.1. Demographic characteristics of subjects

In the current study, 40 samples were taken from MS patients and healthy individuals. The demographic characteristics of the participants in this study are shown in Table 1.

### 3.2. Concentrations of biochemical indicators

Table 2 shows a significant difference in the triglyceride and cholesterol levels of the samples obtained from the MS patients compared to that of the control group. However, no significant difference was observed in the levels of HDL, TAS, and MDA between the two groups. Moreover, the MS group had a lower concentration of triglyceride ($82.76 \pm 27.2$ mg/dL) compared to the healthy group ($148.16 \pm 69.5$ mg/dL). The concentration of cholesterol was lower in the MS group ($162.88 \pm 32.5$ mg/dL) compared to the healthy group ($183.76 \pm 34.8$ mg/dL).

**Table 1**
Demographic characteristics of MS patients and healthy individuals.

| Variable | Case | Control | P-value |
|---|---|---|---|
| **Gender** | | | 0.284 |
| **Male** | 19 | 18 | |
| **Female** | 21 | 22 | |
| **Age/ mean (SD)** | 38.16(7.4) | 40.24 (6.2) | 0.290 |
| **BMI/ mean (SD)** | 23.70 (2.5) | 24.57 (3.1) | 0.283 |
| **mean Duration of diagnosis (month)** | 46.7 | - | - |
| **Plaque in MRI** | | | |
| **Positive** | 29 | - | - |
| **Negative** | 11 | | |
| **Stage of disease** | | | |
| **PPMS** | 2 | | |
| **SPMS** | 3 | | |
| **RRMS** | 28 | | |
| **CIS** | 7 | | |

PPMS: primary progressive multiple sclerosis, SPMS: secondary progressive multiple sclerosis.
RRMS: relapsing-remitting multiple sclerosis, CIS: clinically isolated syndrome sclerosis.
BMI: body mass index, MRI: magnetic resonance imaging.

**Table 2**
Concentrations of biochemical indicators

| Variable | Control group (n = 40) Mean (SD) | Case group (n = 40) Mean (SD) | p-value* |
|---|---|---|---|
| **Triglyceride (mg/dl)** | 148.16 (69.5) | 82.76 (27.2) | <0.0001 |
| **Cholesterol (mg/dl)** | 183.76 (34.8) | 162.88 (32.5) | 0.034 |
| **HDL (mg/dl)** | 47.52 (9.7) | 49.08 (17.8) | 0.703 |
| **TAS (mmol/L)** | 1.40 (0.38) | 1.36 (0.33) | 0.673 |
| **MDA (nmol/ml)** | 1.65 (0.33) | 1.72 (0.41) | 0.528 |

TAS: total antioxidant status, MDA: malondialdehyde, HDL: high-density lipoprotein.
* The p-value was obtained from an independent-samples t-test.

### 3.3. Concentration evaluation of ANT1, ATG5, and Parkin

The serum concentrations of ATG5 and Parkin were significantly higher in the MS patients compared to the healthy group with a p-value of less than 0.0001. However, the serum concentration of ANT1 was lower in the MS patients in comparison to the control group ($p > 0.0001$) (Fig. 1).

### 3.4. Concentration evaluation of TAS, MDA

#### 3.4.1. ROC curve analysis

ROC curve analysis showed the ANT1 level with an area under the curve (AUC) of 0.95 (95% confidence interval (CI): 0.90-0.99, $p < 0.0001$), a sensitivity of 97.5%, and a specificity of 85% with a cutoff point of 2.15 pg/ml. Furthermore, an AUC of 0.99 (95% CI: 0.97-1.0, $p < 0.0001$), a sensitivity 97.5%, and a specificity 97% with a cutoff point of 22.5 ng/ml was obtained for the serum concentration of ATG5. The serum level of Parkin with a cutoff of 20.5 ng/ml indicated a sensitivity of 95%, a specificity of 82%, and an AUC of 0.97 (95% CI: 0.94-1.0, $p < 0.0001$). One may determine that considering a wide range of available tests, ANT1, ATG5, and Parkin can be used to distinguish MS patients from healthy individuals.

### 3.5. Pearson's Correlation Coefficient

Pearson's correlation coefficient was used to evaluate the correlation between the oxidative stress markers and the serum levels of ANT1, Parkin, and ATG5. Based on the obtained results, a weak correlation was observed between the parameters in both MS and control groups, which cannot be considered as statistically significant ($P > 0.05$) (Table 3).

## 4. Discussion

Multiple sclerosis is a demyelinating and inflammatory disease of the central nervous system, affecting many people worldwide. It may cause neurological impairment in patients [15]. The underlying etiology and mechanism of multiple sclerosis are yet to be discovered. The critical role of autophagy and mitophagy in the development and progression of central nervous system diseases has been primarily debated among academics [16,17]. This study revealed that the serum levels of ATG5, as an autophagy factor, and Parkin, as a mitophagy factor, were higher in MS patients compared to the healthy group. Numerous neurodegenerative diseases are associated with an increase in autophagy activity [18]. Previous studies indicated that the number of autophagy and mitophagy-related gene expressions increase in the active phase of the disease [19]. Moreover, research has shown that the expression of ATG5 mRNA in T cells (CD4+, CD8+) is significantly upregulated in MS patients; however, it fails to induce autophagy activities [20].

In addition to autophagy, ATG5 is involved in regulating the immune system, cell death, autoinflammatory diseases, and autoimmune diseases [21]. Overexpression of ATG5 provides survival signals for T cells [11]; these survival signals may increase the inflammatory response in

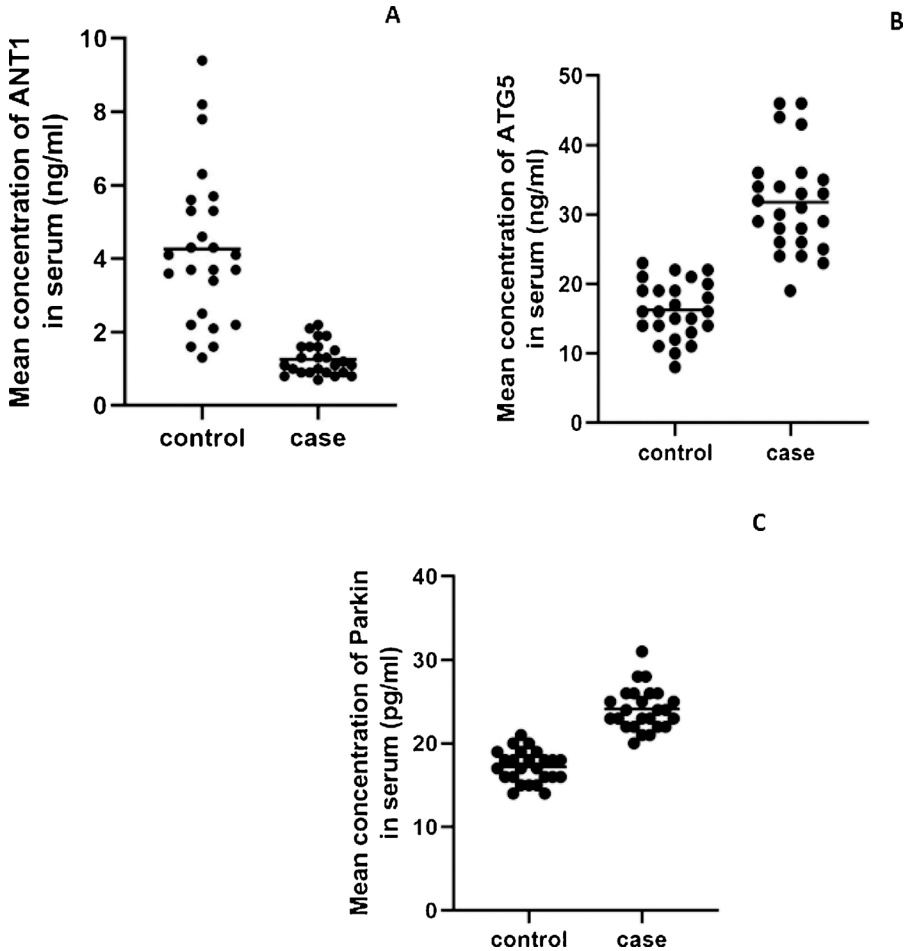


**Fig. 1.** Comparison of the serum concentration of ANT1 (A), ATG5 (B) and Parkin (C) in the case and control groups. Statistical analysis was conducted by the independent sample t-test ($P < 0.0001$). ATG5; autophagy protein 5, ANT1; adenine nucleotide translocase-1.

**Table 3**
The correlation between the oxidative stress markers and the serum levels of ANT1, Parkin, and ATG5.

| Variable | Case group | | | | Control group | | | |
|---|---|---|---|---|---|---|---|---|
| | MDA | | TAS | | MDA | | TAS | |
| | **r*** | **P value** | **r** | **P value** | **r** | **P value** | **r** | **P-value** |
| **ANT1** | −0.095 | 0.651 | 0.075 | 0.723 | 0.042 | 0.841 | 0.183 | 0.382 |
| **ATG5** | −0.343 | 0.093 | 0.037 | 0.861 | 0.287 | 0.164 | 0.090 | 0.668 |
| **Parkin** | 0.190 | 0.362 | 0.121 | 0.564 | 0.227 | 0.275 | 0.207 | 0.320 |

Pearson's correlation coefficient was used; the correlation was defined as significant at $P < 0.05$. ANT1: adenine nucleotide translocase 1, ATG5: autophagy-related 5, TAS: Total antioxidant status.
* Correlation coefficient

the central nervous system of MS patients, which can lead to further problems.

In a 2019 study by Massimiliano et al., it was indicated that the serum levels of Parkin and ATG5 are lower in patients with Alzheimer's, mild cognitive impairment (MCI), and mixed dementia (MI) in comparison to the healthy group [22]. These results were in contrast with our findings, where the serum levels of Parkin and ATG5 were higher in MS patients compared to the healthy group. While the serum levels of ATG5 and Parkin were higher in the MS group, the present study results indicated a low concentration of ANT1 in MS patients; this is probably due to the inhibitory effects of the nuclear factor-kappa β (NF-Kβ). NF-Kβ plays a significant role in regulating the immune response; namely, incorrect regulation of this protein can lead to inflammatory diseases, including rheumatoid arthritis, atherosclerosis, asthma, and multiple sclerosis [23]. NF-Kβ downregulates the transcription of the ANT1 gene and ATP production level, leading to the degradation of CNS [12,24].

In the study of "Nutrition Status and Antioxidant Capacity in Multiple Sclerosis Patients" by Armon-Omer, it was indicated that MS patients have a statistically significant lower intake of various nutritional components [25]. This, in turn, can affect the concentration of many biochemical indicators. As shown in Table 2, the serum concentrations of triglyceride and cholesterol are significantly lower in MS patients than in the control group (Figs. 2 and 3).

Three of the well-studied biological markers of oxidative stress are the advanced oxidation protein product (AOPP) as the protein oxidation marker, the thiobarbituric acid reacting substances (TBARS) as the lipid peroxidation, and the advanced glycation end-products (AGEs) as the carbonyl stress [26].

While oxidative stress increases considerably in the progressive stages of multiple sclerosis, the severity of oxidative stress in the early

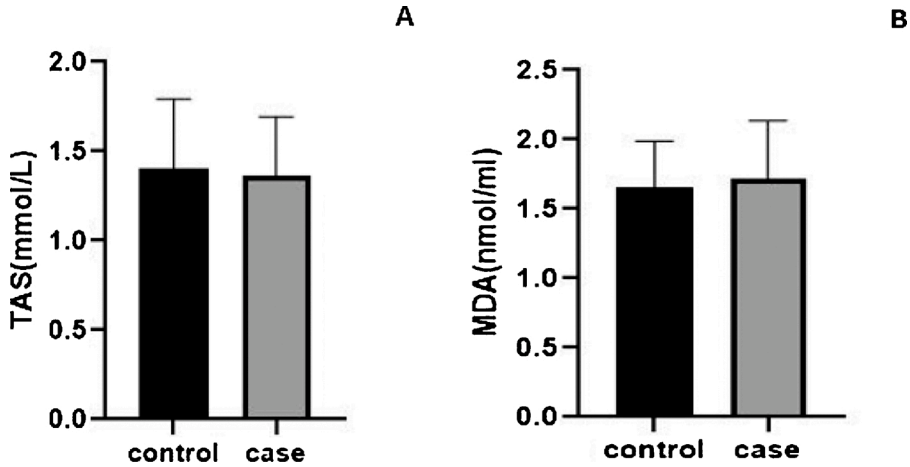


**Fig. 2.** Comparison of the serum concentration of TAS (A(and MDA)B(in the case and control groups. The mean level of oxidative stress markers did not show a significant difference between the case and control groups.)P > 0.05(.TAS: Total antioxidant status.

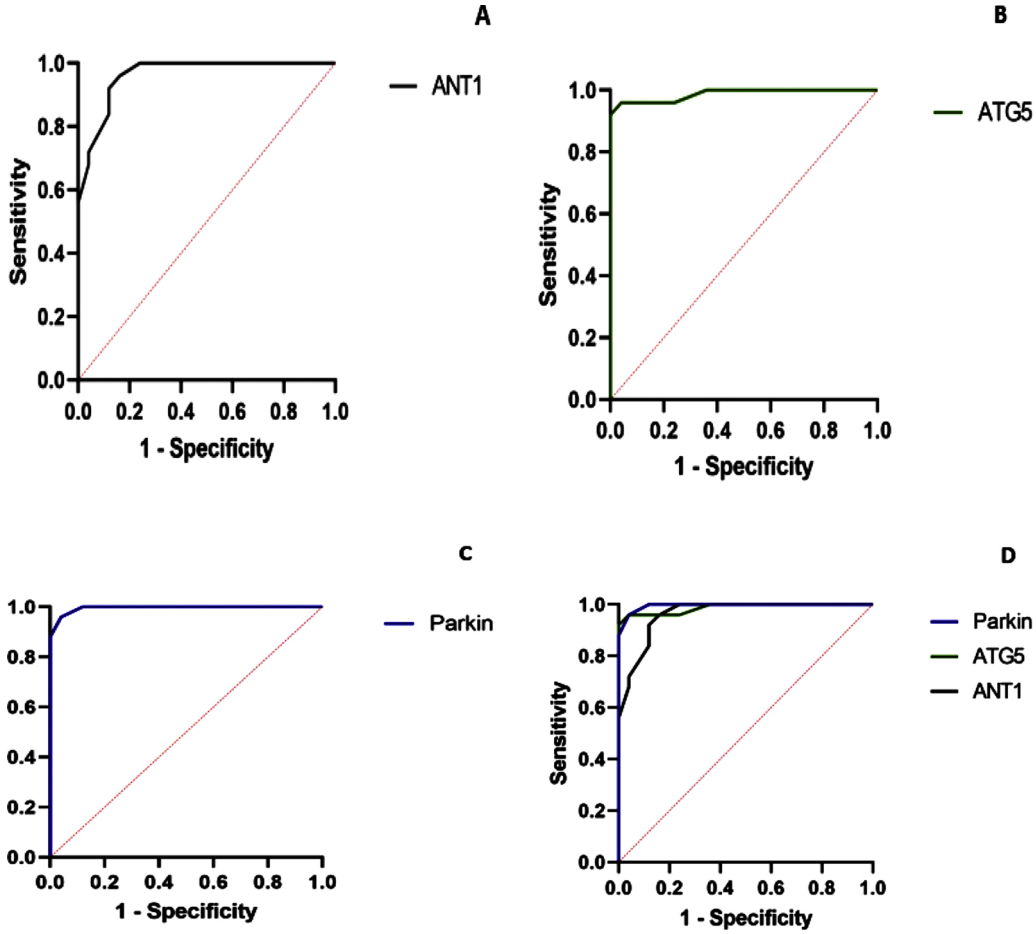


**Fig. 3.** ROC curve analysis for serum concentrations of ANT1 (A), ATG5 (B), Parkin (C), and a combination of three factors (D). ANT1: adenine nucleotide transporter 1, ATG5: autophagy-related 5.

stages of the disease depends primarily on the inflammatory response. On the other hand, lipid peroxidation, measured by the generation of TBARS in RRMS patients, is not significantly different from the healthy group [27,28]. Furthermore, no significant correlation was observed between oxidative stress and autophagy and mitophagy factors, which may be due to the low level of oxidative stress at the initial stages of the disease (RRMS). It was also revealed that MS patients have a lower level of glutathione (GSH) than the healthy group. However, this difference between the RRMS patients and the control group was not statistically significant [29]. Finally, oxidative stress and lipid peroxidation were significantly higher in PMS patients than the control group [30].

Oxidative stress plays an essential role in the severity of MS; moreover, the evidence suggests that antioxidants can prevent the disease's progression. Also, free radicals can activate the NF-kβ pathway and enhance inflammation response. The treatment of MS with antioxidant enzymes in mouse models was effective in decreasing the severity of the

disease [31,32].

## 5. Conclusion

Our results indicated that the serum levels of autophagy and mitophagy factors were higher in MS patients, and the serum level of ANT1 was lower than the control group. These differences between the case and control groups were statistically significant. Based on the present study's findings, the serum concentrations of TAC and MDA were lower and higher in MS patients than the control group, respectively. However, no significant correlation was observed between the two groups. Furthermore, no significant association was found between oxidative stress and autophagy or mitophagy factors since oxidative stress is low in the early stages of the disease. It is recommended that studies be performed on a large population of MS patients to accurately determine these parameters' potential in diagnosing or evaluating treatment.

## Declaration of Competing Interest

The authors declare that they have no conflict of interest.

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## CRediT authorship contribution statement

**Omid joodi khanghah:** Methodology, Writing-original draft. **Alireza Nourazarian:** Conceptualization, Methodology, Writing-review & editing. **Fatemeh Khaki-Khatibi:** Methodology, Formal analysis, Data curation. **Masoud Nikanfar:** Methodology, Formal analysis, Data curation. **Delara Laghousi:** Data curation, Writing-review & editing. **Amir Mansour Vatankhah:** Conceptualization, Methodology. **Sepideh Moharami:** Data curation, Methodology.

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